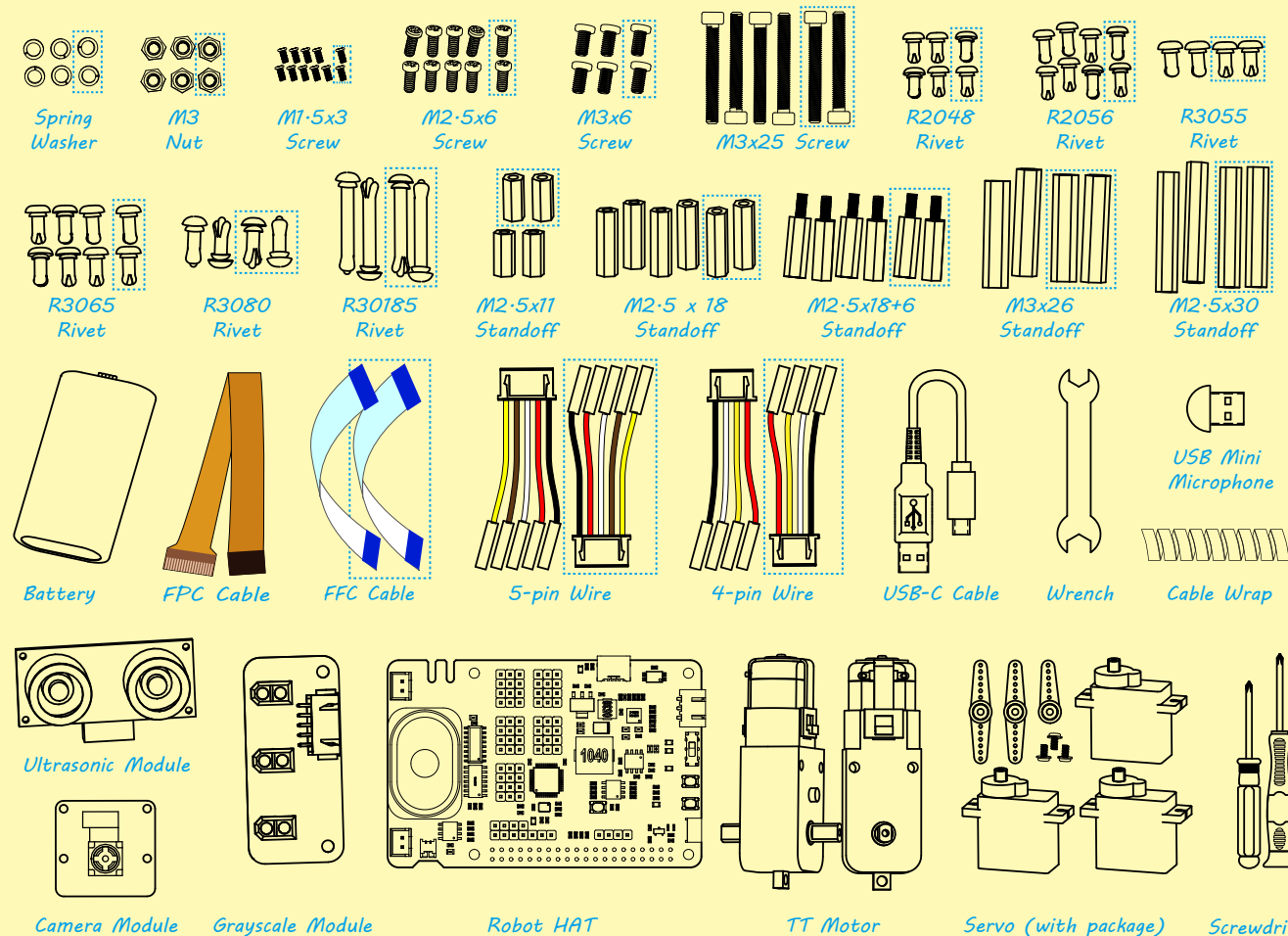
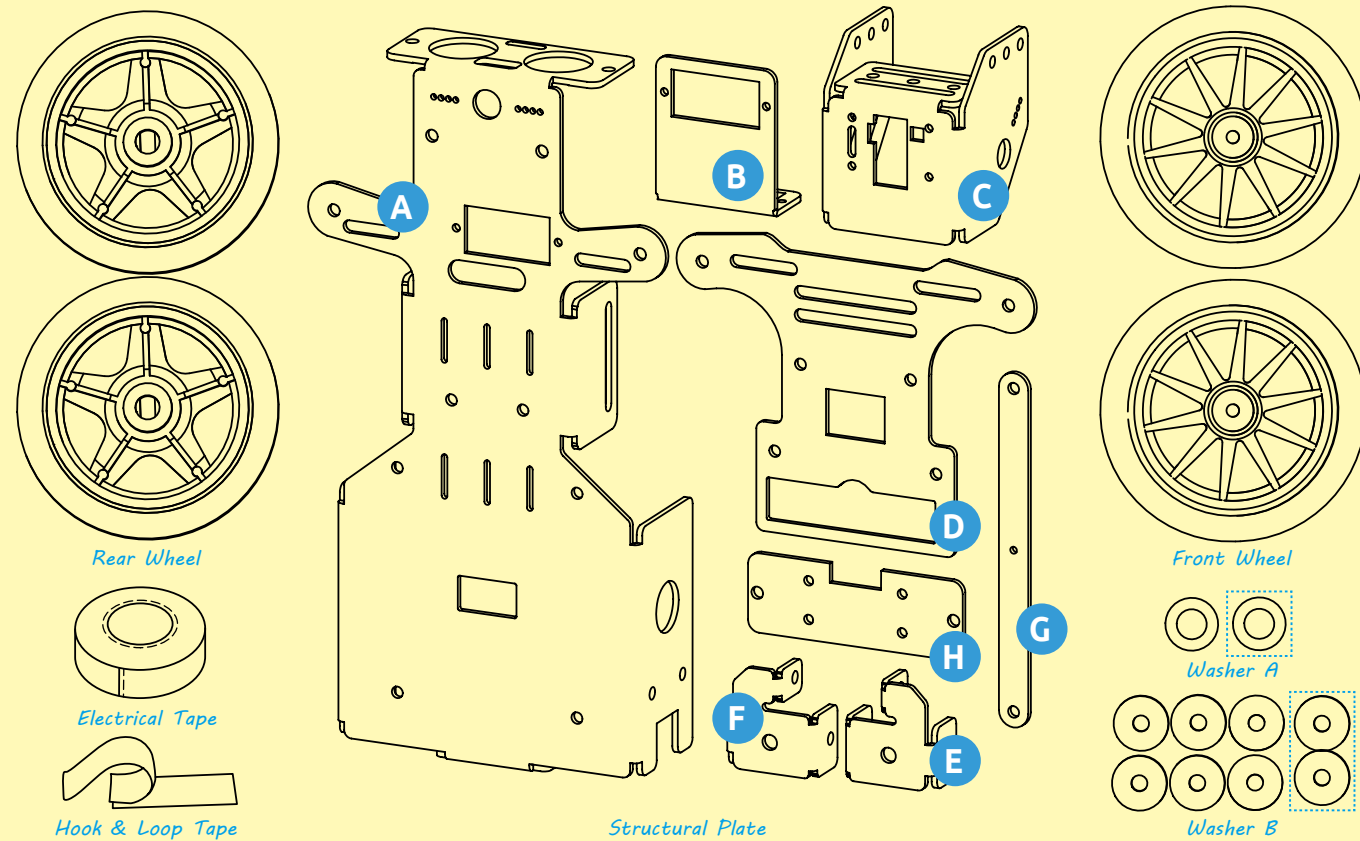


SUNFOUNDER PICAR-X KIT

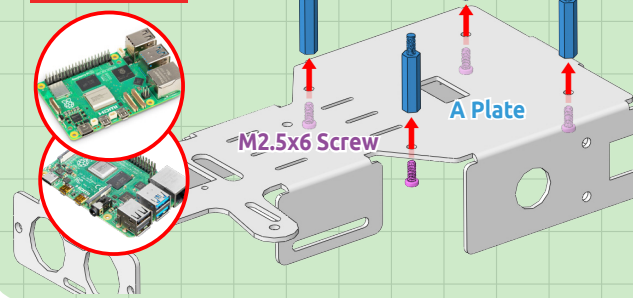
Get tutorial at: <https://picar-x-v20.rtf.d.io>



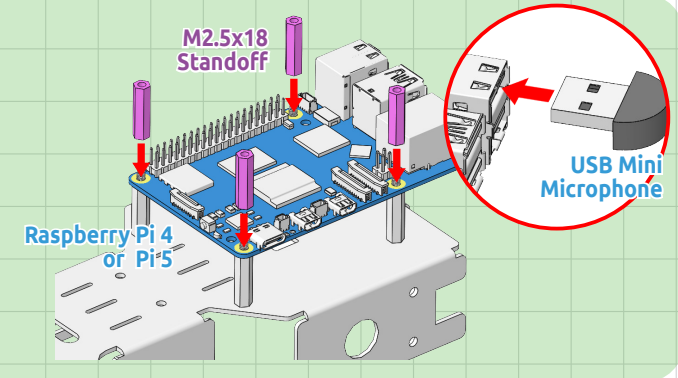
Note: The items framed are for backup.

ASSEMBLE

Raspberry Pi 4
Raspberry Pi 5

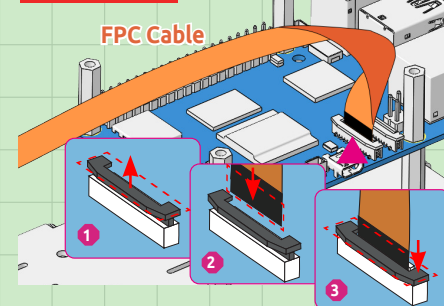


Step 1
First, screw 4 nylon standoffs into the car body.



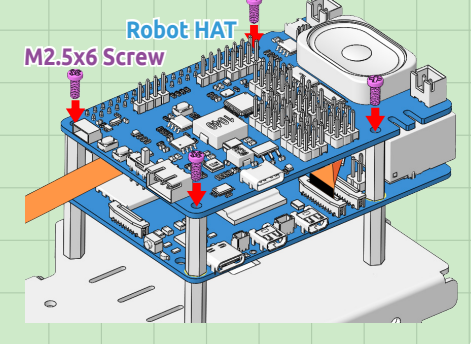
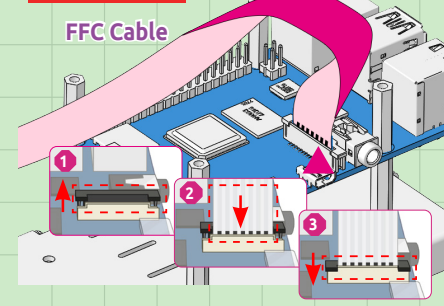
Step 2
Place the Raspberry Pi onto the standoffs, and secure it with additional nylon standoffs.

Raspberry Pi 5



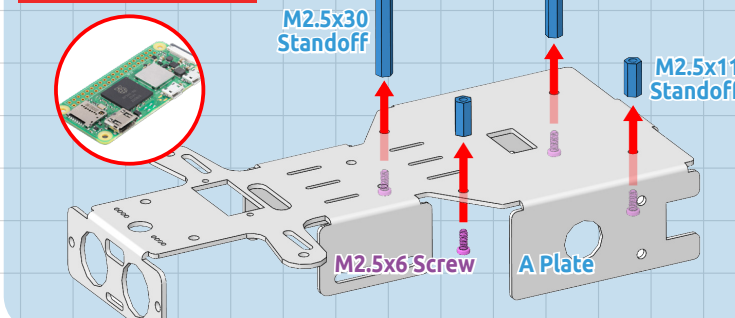
Step 3
Lift the black tab on the Raspberry Pi's camera connector, insert the FFC/FPC cable and press down to fasten it.

Raspberry Pi 4

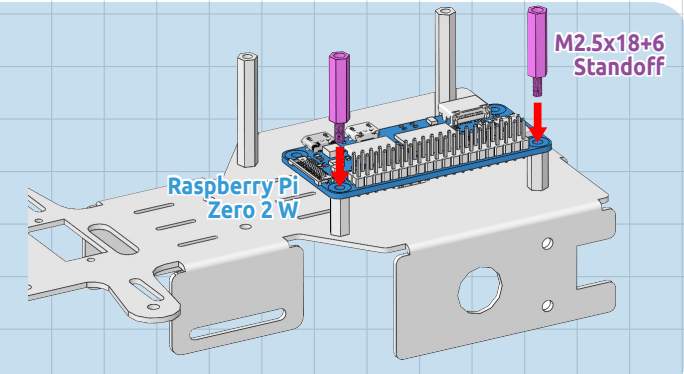


Step 4
Insert the Robot HAT into the Raspberry Pi and secure it with screws.

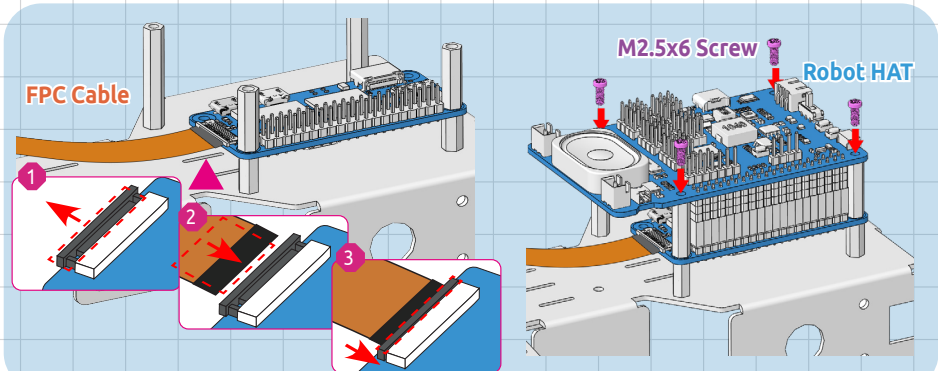
Raspberry Pi Zero 2 W



Step 1
Install two sets of different sized nylon standoffs onto the car body, one set will be used to secure the Raspberry Pi Zero 2 W.

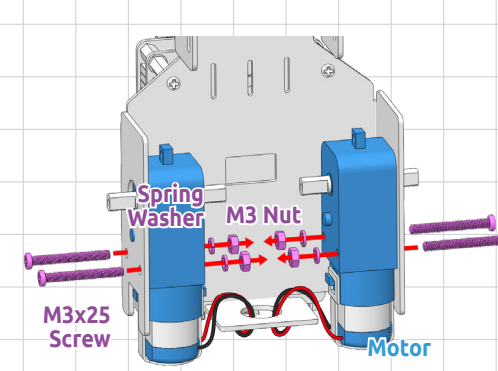


Step 2
Place the Raspberry Pi Zero 2 W onto the standoffs, and secure it with additional nylon standoffs.

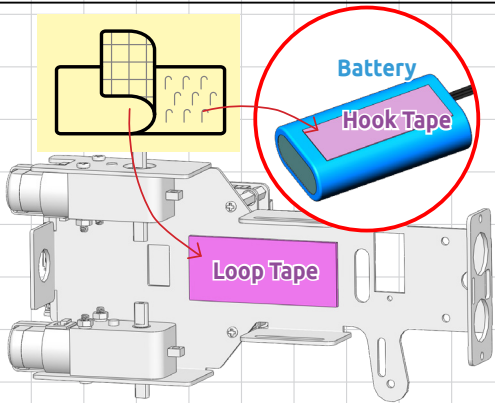


Step 3
Pull out the tab on the camera connector, insert the FPC cable, and then press it back in to secure.

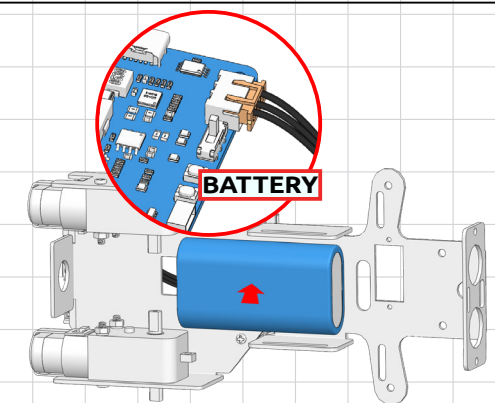
Step 4
Insert the Robot HAT into the Raspberry Pi Zero and secure it with screws.



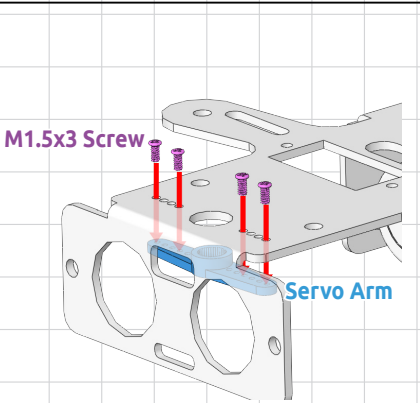
Step 5
Install two motors at the rear of the car and ensure the motor wires face inward.



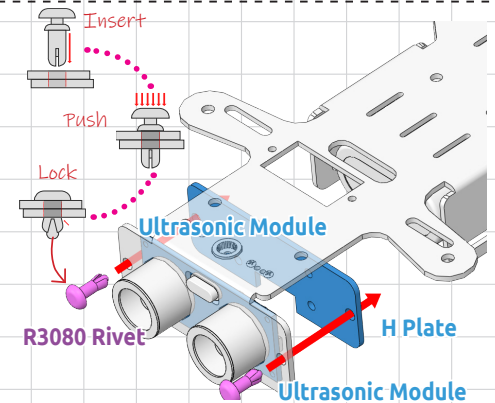
Step 6
Attach the hook tape to the battery and the loop tape to the underside of the car.



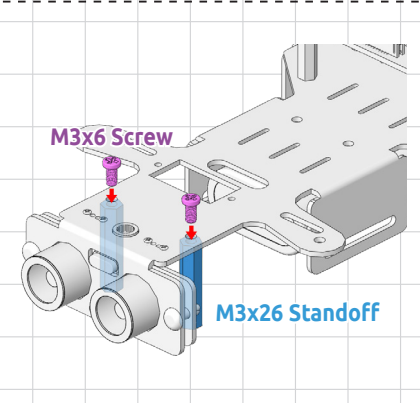
Step 7
You can now secure the battery in place. Don't forget to connect the battery wire!



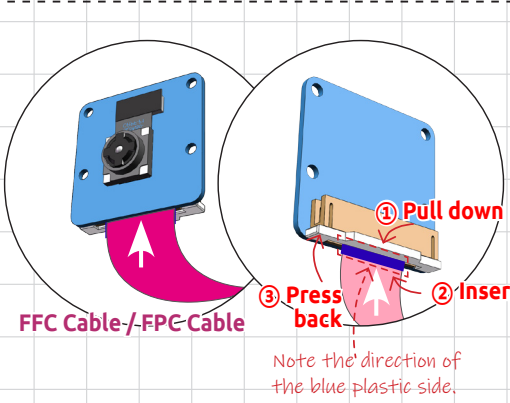
Step 8
Secure a servo arm at the front of the car to hold the pan-tilt.



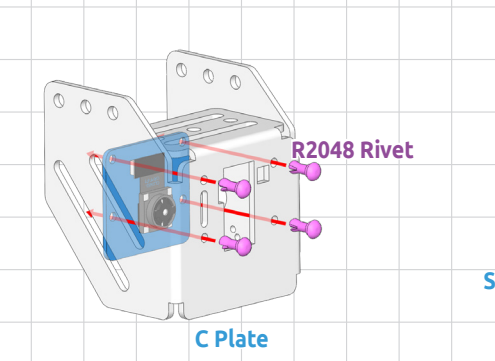
Step 9
Using rivets and a plate, secure the ultrasonic module to the front of the car.



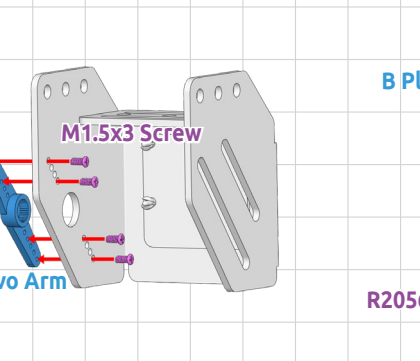
Step 10
Fasten 2 standoffs to the bottom of the car with screws.



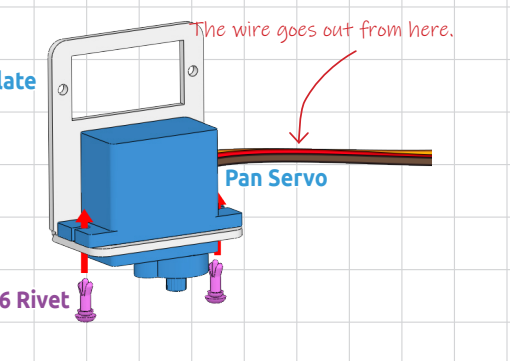
Step 11
Secure the other end of the FFC or FPC cable to the camera module.



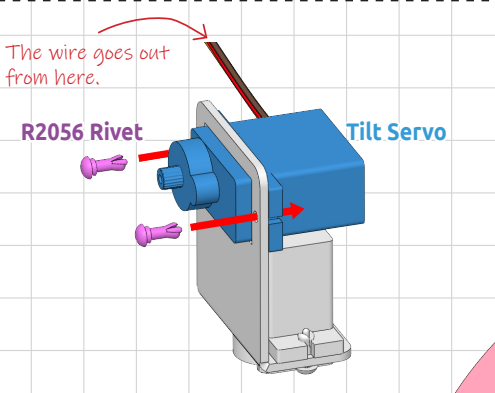
Step 12
Then, mount the camera module onto the C plate.



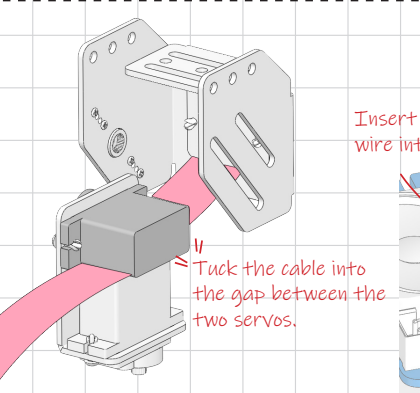
Step 13
Additionally, mount a servo arm on the C plate.



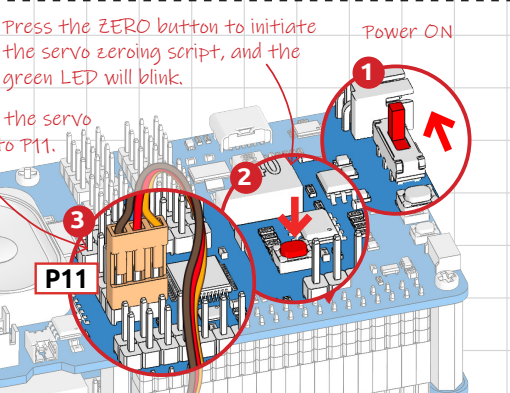
Step 14
Install a servo on the B plate, marking it as "Pan". Pay attention to the direction of the servo wire.



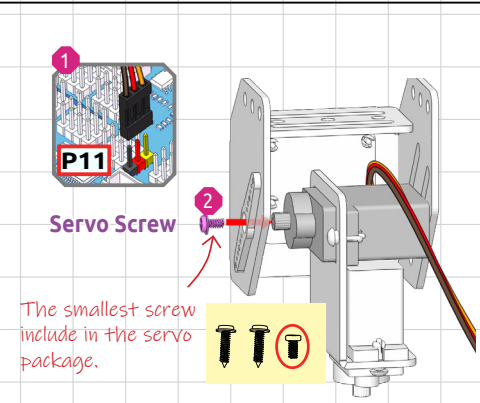
Step 15
Attach another servo on the B plate, marked as "Tilt". Note the direction of the wire as well.



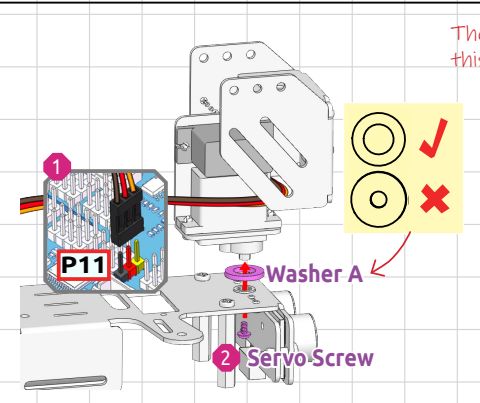
Step 16
Pass the FFC cable (or FPC cable) of the camera through the gap between the Pan and Tilt servos on the B plate.



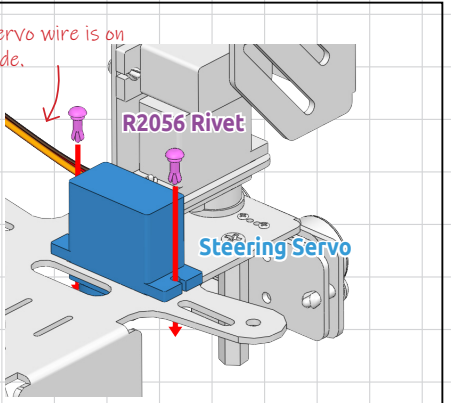
Step 17
Power ON, press the Zero button. Next, before attaching the servo arms, each servo must be connected to P11 to zero its angle.



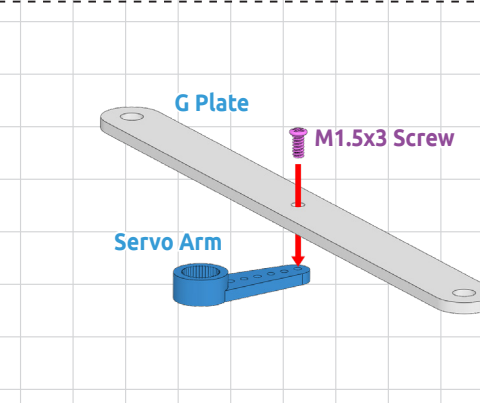
Step 18
Adjust servo angle to the middle position before securing the servo and servo arm.



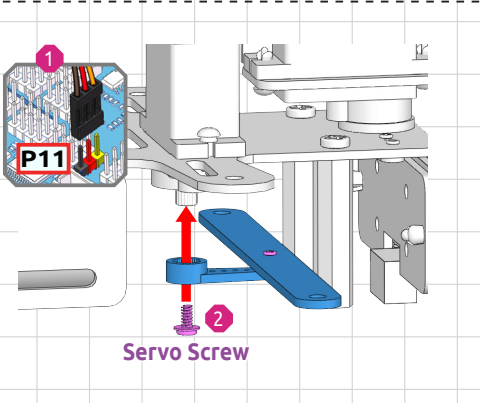
Step 19
Similarly, adjust the servo angle prior to securing the servo and servo arm.



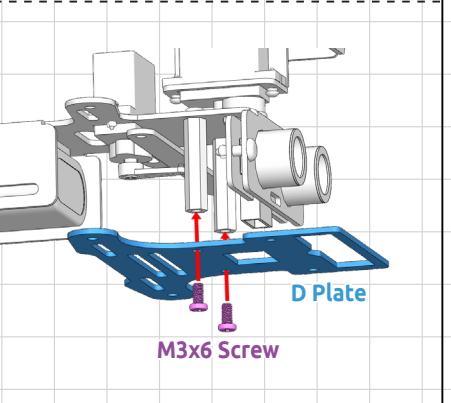
Step 20
Secure the remaining servo and label it as "Steering". Pay attention to the direction of the servo wire.



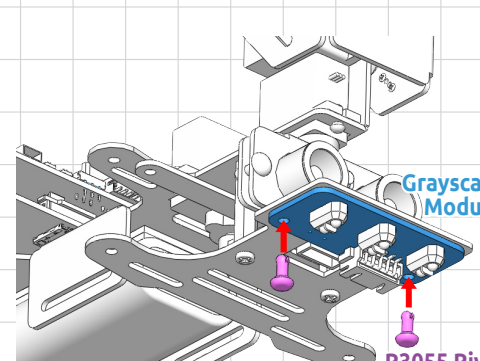
Step 21
Secure a servo arm onto the G plate using a screw, ensuring not to overtighten it, allowing for **free rotation** between them.



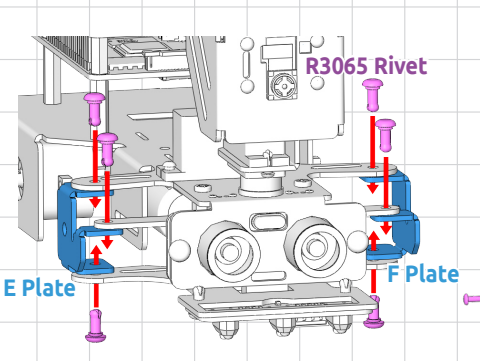
Step 22
Adjust the angle of the "Steering" servo, insert the previously mentioned servo arm, and secure them using a servo screw.



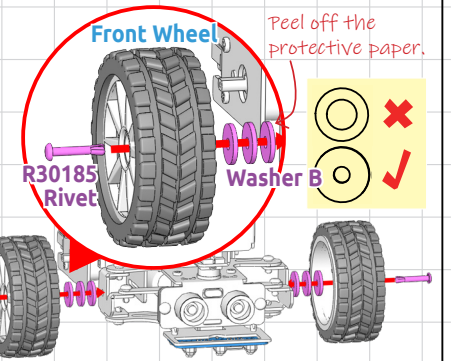
Step 23
Secure the D plate in place using screws.



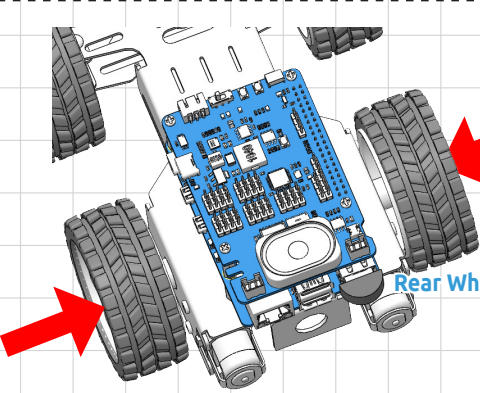
Step 24
Next, use rivets to secure the grayscale module onto the D plate.



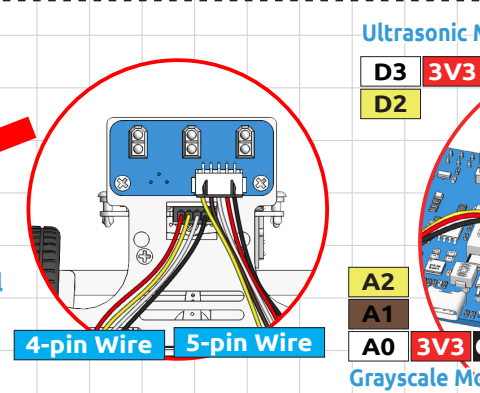
Step 25
Attach the E plate on the right side and the F plate on the left side for the front wheels.



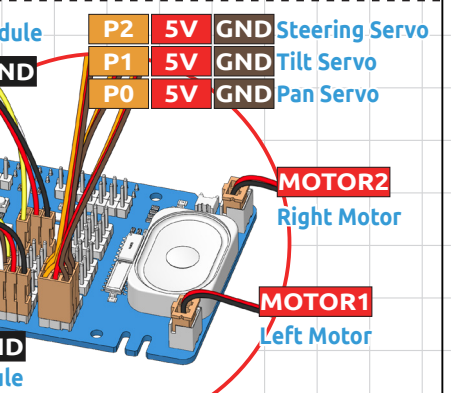
Step 26
You can now install the front wheels.



Step 27
Insert the rear wheels onto the motor shafts.



Step 28
Plug in the 4-pin wire into the ultrasonic module first, and then insert the 5-pin wire for the grayscale module.



Step 29
As per the diagram, insert all the wires into their designated ports on the Robot HAT to complete the assembly.